How Much of a Time Series Survives Its Image? A Cross-Domain Comparative Analysis of the Reconstruction Capacity of Image-Based Representations

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Abstract

Image-based representations have been widely studied for time-series classification, while recent work has also demonstrated their use for image-based synthesis and imputation. Yet successful generation does not establish what different encodings preserve or whether their images can be reliably decoded into signals. We investigate this question for binary, unsigned and signed recurrence plots (RP), Gramian angular summation and difference fields (GASF/GADF), and Markov transition fields (MTF), using the invertible-by-design XIRP as a reference. Alongside established inverses, we introduce an image-only geometric decoder for signed RP, a restricted-range GADF decoder, and an approximate MTF reconstruction procedure.

Across seven datasets from six domains, we evaluate temporal and spectral fidelity, robustness, computational cost and downstream classification utility, distinguishing encoding losses from 8-bit storage effects. Under stated conditions, signed RP, GASF, restricted-range GADF and XIRP allow numerically exact reconstruction from floating-point matrices; storage introduces addi-

tional error, particularly for ill-conditioned RP states. Unsigned RP retains a global-reflection ambiguity, whereas binary RP and MTF are intrinsically many-to-one.

These findings establish a reproducible comparison of reconstruction capabilities and limitations, informing future research on generating time series through image-domain models.
Keywords: Time-series imaging, Reconstruction, Recurrence plots, Gramian angular fields, Markov transition fields, Invertibility, Comparative analysis

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1. Introduction

Image-based representations transform time series into structured matrices that can be processed using techniques originally developed for images. Recurrence plots (RP), Gramian angular fields (GAF) and Markov transition fields (MTF) have been widely explored for time-series classification and related tasks [1, 2, 5, 6]. In classification, a representation need only preserve information relevant to the target task; reconstructing the original waveform is not necessarily required. However, applications that aim to recover or synthesise temporal signals introduce a different requirement: the encoded images must retain sufficient information to reconstruct meaningful time series.

Recent research has demonstrated the feasibility of exploiting image-based generative models for temporal data. Huang et al. [3] used Gramian angular representations with generative adversarial networks for traffic-data imputation. More recently, Hellermann and Lessmann [4] introduced the extended intertemporal return plot (XIRP), an invertible-by-design represen-

tation for time-series generation through image-based GANs. Their results demonstrated competitive predictive utility against specialised generative approaches and highlighted the importance of the inversion procedure. These developments establish image-based time-series generation as a promising research direction. Nevertheless, successful synthesis using particular representations does not establish the reconstruction capabilities of other encodings, nor the conditions under which their information can be recovered.

Understanding these properties requires distinguishing three aspects: the information intrinsically preserved by an encoding, the ability of a decoder to recover that information, and the stability of reconstruction after image storage or perturbation. This distinction is particularly relevant for representations whose inversion is ambiguous or has received comparatively limited attention. Moreover, accurately reconstructing an image produced by a known encoder does not guarantee that an arbitrary image generated by a learned model represents a valid time series. Characterising reconstruction is therefore a necessary preliminary investigation, rather than a demonstration of generative performance.

In this work, we establish a common framework for evaluating the reconstruction capabilities of image-based representations built from relationships between temporal samples or states. We examine binary, continuous unsigned and signed RP; Gramian angular summation and difference fields (GASF/GADF); and MTF. XIRP is included as an invertible-by-design reference developed for generative applications. Time-frequency and time-scale representations, such as spectrograms and scalograms, are outside the scope of this study.

We combine established reconstruction methods with new image-only procedures: a geometric decoder for signed RP, an anchored decoder for GADF under restricted angular normalisation, and an approximate representative-trajectory decoder for MTF. For recurrence plots, we distinguish the contribution of the encoding from that of the reconstruction algorithm through comparisons with the approaches of Hirata et al. [12] and Sipers et al. [14]. All methods are evaluated under a common 8-bit image-storage protocol, reconstructing normalised signals from the stored images alone, with shared calibration used only to recover physical units (Figure 1).

We address four research questions:

- *RQ1 — Recoverability.* What information can be recovered from each representation, in both the time and frequency domains, and how much is lost through encoding and 8-bit storage?
- *RQ2 — Reconstruction methods.* Under which conditions are the proposed decoders exact, and how do the encoding and decoding stages separately influence recurrence-plot reconstruction?
- *RQ3 — Robustness and stability.* How do image perturbations, quantisation and signal characteristics, including sampling rate and state-space conditioning, affect reconstruction fidelity?
- *RQ4 — Computational and downstream utility.* What are the computational costs of reconstruction, and to what extent do reconstructed signals preserve the performance of classifiers trained on original data?

The main contributions of this study are:

- *A reproducible reconstruction benchmark.* We systematically evaluate temporal image representations across seven public datasets from six domains, using a common protocol covering temporal and spectral fidelity, robustness, computational cost and downstream classification utility. This allows intrinsic representation losses to be distinguished from storage-related errors.

- *New reconstruction procedures and theoretical characterisation.* We introduce a geometric image-only decoder for signed RP, together with sufficient conditions for exact recovery, an anchored decoder for restricted-range GADF, and an approximate reconstruction procedure for MTF. Controlled RP comparisons further separate the effects of directional information, distance preservation and decoder design.

- *An empirical foundation for image-based time-series generation.* We identify which representations admit reliable reconstruction, the ambiguities and limitations they retain, and the conditions affecting reconstruction stability. These findings provide a methodological basis for future generative studies, without assuming that reconstruction fidelity alone determines the quality of synthetic time series.

2. Related work

Temporal image encodings.. Recurrence plots (RP) visualise recurrences in an embedded state space, conventionally through a binary neighbourhood relation [5, 6]. Continuous RP variants preserve state distances; the modified signed RP of Lu and Tong additionally provides directional information [11].

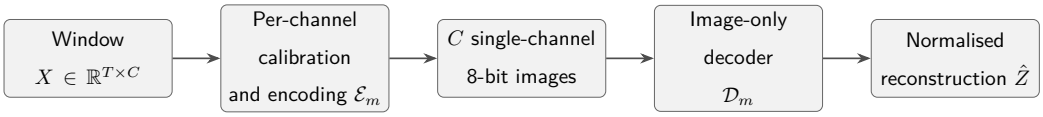


Figure 1: Controlled reconstruction protocol. Each channel is calibrated, encoded and stored as an 8-bit image. The decoder recovers a normalised trajectory from the image alone; shared per-channel calibration is used only to return to physical units. Decoding before quantisation isolates reconstruction error from storage error.

Wang and Oates introduced Gramian angular summation and difference fields (GASF/GADF) and Markov transition fields (MTF) as image representations for classification and imputation [1, 2]. Their subsequent use in activity recognition and anomaly detection illustrates the value of these encodings as predictive features [7, 8]. Predictive usefulness, however, does not establish whether the original waveform can be recovered.

Reconstruction from encoded images.. The inverse problem has been investigated most directly for RP. Hirata et al. reconstruct state-space geometry from binary recurrence information, using graph-based distances and multidimensional scaling [12, 23]. For unthresholded RP, Sipers et al. establish identifiability conditions up to mean and global sign and develop a spectral reconstruction method, including experiments on disturbed and multilevel images [13, 14]. These results concern unsigned recurrence information and therefore do not resolve orientation using the directional content of a signed RP. For GASF, the diagonal inverse on a nonnegative normalisation is already known and has supported image-based imputation [1, 9]. By contrast, GADF retains angular differences but not an immediately available absolute reference, and MTF quantisation discards within-state amplitudes.

Image-domain generation and the remaining question.. Recent work has extended temporal imaging beyond prediction: GAN-based image generation has been used for traffic-data imputation [3], while Hellermann and Lessmann introduced XIRP, a deliberately invertible encoding, for synthetic time-series generation with image-based GANs [4]. XIRP provides a useful reference because its diagonal explicitly stores the normalised trajectory. These contributions motivate a controlled study of what other images determine, how decoders use that information, and how 8-bit storage and perturbations affect reconstruction. For RP, we also separate encoding effects from decoder effects.

3. Mathematical formulation

3.1. Reconstruction problem and information retained

Let $x = (x_1, \ldots, x_T)$ denote a non-constant window of one signal channel. For representation m, the calibrated series $z_m = \mathcal{N}_m(x; x_{\min}, x_{\max})$ lies in a predefined normalised domain. It is encoded as a matrix, stored as an 8-bit image, and decoded according to

$$I_m = \mathcal{Q}_m\Big(\mathcal{E}_m(z_m)\Big), \qquad \hat{z}_m = \mathcal{D}_m(I_m), \qquad \hat{x} = \mathcal{N}_m^{-1}(\hat{z}_m; x_{\min}, x_{\max}).$$

Here $\mathcal{Q}_m$ is a predefined storage *rule*, possibly with a per-image affine range. The term *image-only* refers to $\hat{z}_m$: decoding never accesses the original window or window-specific side information. The shared per-channel pair $(x_{\min}, x_{\max})$ is required only to express $\hat{x}$ in physical units, a distinct issue for future image-based generators (Figure 1).

For readability, encoding formulas below use x to denote the calibrated signal z_m. Three properties must be distinguished. *Identifiability* concerns

whether two admissible normalised signals can produce the same encoded matrix, possibly up to a specified symmetry. *Decoder exactness* concerns whether $\mathcal{D}_m \circ \mathcal{E}_m$ recovers the original normalised signal under stated hypotheses, before storage. *Stability* concerns how errors in the image, including quantisation, propagate to the decoded signal. A many-to-one encoder cannot admit a universally exact inverse, irrespective of decoder design. Conversely, a mathematically identifiable signal may still be reconstructed inaccurately from an 8-bit image if the inverse is poorly conditioned.

3.2. Recurrence plots

For all RP variants we use adjacent two-sample states,

$$s_i = (x_i, x_{i+1}) \in \mathbb{R}^2, \qquad i = 1, \ldots, T-1,$$

that is, a two-dimensional delay embedding with unit delay [10]. The variants differ in how much information they retain about pairwise state displacements.

The *binary RP* records neighbourhood membership, $B_{ij} = \mathbb{1}[\|s_i - s_j\|_2 < \varepsilon]$. Thresholding discards the underlying distances, making the encoding many-to-one [5, 6]. The *continuous unsigned RP* stores $D_{ij} = \|s_i - s_j\|_2$, scaled by $\max_{ij} D_{ij}$. As a Euclidean distance matrix, D determines the embedded states up to translation and an orthogonal transformation. In particular, reflecting all signal values leaves D unchanged. Recovery up to global reflection therefore requires additional non-degeneracy assumptions; the unsigned image cannot determine the absolute orientation by itself [13].

The *signed RP* of Lu and Tong [11] associates a directional label with each distance. With $v = (1, 1)/\sqrt{2}$ and $\hat{u}_{ij} = (s_i - s_j)/\|s_i - s_j\|_2$ for nonzero

displacements, its encoding is

$$R_{ij} = \sigma_{ij}\|s_i - s_j\|_2, \qquad \sigma_{ij} = \begin{cases} +1, & \|s_i - s_j\|_2 = 0 \text{ or } \hat{u}_{ij} \cdot v \geq \cos(\pi/4), \\ -1, & \text{otherwise.} \end{cases} \tag{1}$$

Its magnitude recovers the unsigned matrix, $|R| = D$. Directional information can additionally distinguish a signal from its global reflection: a pair is *sign-informative* when its displacement lies within $\pi/4$ of v or $-v$, in which case $\sigma_{ij} = -\sigma_{ji}$. Other nonzero pairs receive the same negative sign in both directions. Thus the signed RP adds information that is absent from the distance matrix, but this alone does not guarantee unique reconstruction. Its per-window affine image storage retains the quantised zero level through the diagonal (Section 4.1).

For exact-input recovery, three issues must be resolved: the embedded geometry must be recoverable from distances; consecutive embedded states must determine a consistent one-dimensional trajectory; and the signed information must distinguish the remaining reflection.

These requirements are not always met: periodic or nearly alternating signals may violate geometric uniqueness or the sign condition. Thus recovering distances alone need not identify the original series. In the benchmark, all 61 083 channel series have a sign-informative pair (minimum share 6%), and the floating-point decoder recovered every one of them to numerical precision (Section 6.1).

3.3. Gramian angular fields

GASF and GADF encode pairwise angular relations [1]. A normalised sample $\tilde{x}_i \in [-1, 1]$ is mapped to $\phi_i = \arccos \tilde{x}_i$, giving

$$G^{\mathrm{S}}_{ij} = \cos(\phi_i + \phi_j), \qquad G^{\mathrm{D}}_{ij} = \sin(\phi_i - \phi_j).$$

Their inverse properties differ substantially. For *GASF*, the diagonal is $G^{\mathrm{S}}_{ii} = \cos(2\phi_i)$. It is ambiguous over $[-1, 1]$ but injective when samples are normalised to $[0, 1]$, as in our experiments; there $\tilde{x}_i = \cos\!\left(\frac{1}{2} \arccos G^{\mathrm{S}}_{ii}\right)$. Thus GASF has a direct image-only inverse under the stated normalisation.

For *GADF*, the diagonal is identically zero and only relative angular differences are available. Recovering the series requires both a unique branch of arcsin and an angular reference. With per-window normalisation to $[-\alpha, \alpha]$, the maximum angle separation is

$$\max_{i,j} |\phi_i - \phi_j| = \pi - 2 \arccos \alpha < \frac{\pi}{2} \qquad \text{if} \quad \alpha < \frac{1}{\sqrt{2}}.$$

We use $\alpha = 0.7$, so angular differences remain on the principal branch. Because every non-constant calibrated window reaches both endpoints, the global offset can then be fixed using the image itself (Section 4.4). This conditional inverse does not extend to arbitrary GADF normalisations: outside the restricted interval, branch ambiguities may remain.

3.4. Markov transition fields

MTF first assigns each sample to a quantile state $q_t \in \{1, \ldots, Q\}$ and estimates the transition probabilities $W_{pq} = P(q_{t+1} = q \mid q_t = p)$ [1]. The image is then defined by $M_{ab} = W_{q_a q_b}$. It retains information about the temporal arrangement of discrete states but discards within-state amplitudes;

moreover, numerical state ordering cannot generally be inferred uniquely from the image. Different waveforms can therefore yield the same MTF. In this setting reconstruction is necessarily approximate: the decoder selects a representative trajectory rather than recovering a uniquely determined original.

3.5. Extended intertemporal return plot

XIRP was proposed specifically for image-based time-series generation [4]. For a strictly positive normalised series it combines intertemporal log returns with an explicit record of the sample values:

$$R_{ij} = \begin{cases} \log(x_i/x_j), & i \neq j, \\ x_i, & i = j. \end{cases}$$

The diagonal thus provides an exact inverse before storage, while every off-diagonal column supplies additional estimates through $x_i = x_j \exp(R_{ij})$. We map each channel to $[1, 2]$ with the common calibration and include XIRP as an *invertible-by-design reference*, not as another conventional classification encoding.

Table 1 summarises these recoverability properties; Figure 2 displays the actual stored images for representative signal windows.

4. Decoders

The previous section describes what each encoding retains; here we describe how that information is used to reconstruct a trajectory. All methods operate on the stored image. Published methods are distinguished from

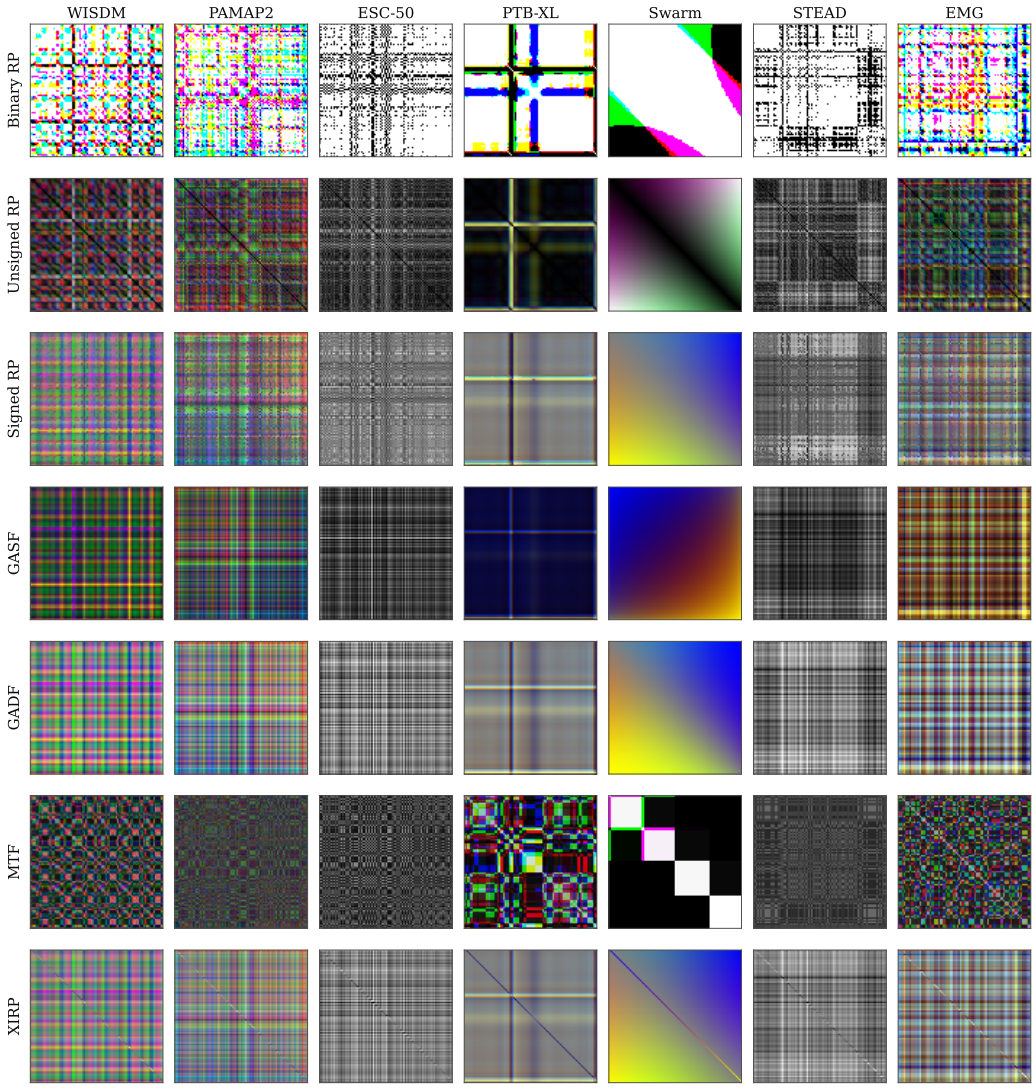


Figure 2: Representative 8-bit encoding images ($T = 128$; one window per dataset). Each channel is stored as its own single-channel image; for display, the first three channels are shown as the red, green and blue planes of one picture, and the single-channel ESC-50 appears in grey. The binary RP uses a recurrence rate of 0.75. The red (first) channel is decoded in Figure 3.

Table 1: Information retained by each encoding and reconstruction approaches evaluated. Exactness refers to unquantised matrices under the stated conditions; decoders operate on normalised signals using the image alone.

Encoding	Image determines	Exact recovery	Decoders (origin)
Binary RP	Recurrence neighbourhoods	No (many-to-one)	Hirata's graph method, with our sample reconstruction
Unsigned RP	State geometry	Up to reflection under non-degeneracy	Hirata after thresholding [12]; Sipers on circular RP [14]; geometric method without sign (this work)
Signed RP	Geometry and directional information	Conditional	Hirata and Sipers with adapted sign selection; proposed geometric decoder
GASF	Angular sums	Yes on $[0, 1]$	Diagonal inverse [1]
GADF	Angular differences	Yes when $\alpha < 1/\sqrt{2}$ with bounded normalisation	Restricted-branch and boundary-anchor decoder (this work)
MTF	Discrete-state transitions	No (many-to-one)	Spectrally ordered representative (this work)
XIRP	Returns and explicit diagonal values	Yes	Inversion by mean [4]

our adaptations, and the RP experiments compare both changes of encoded information and changes of decoder.

4.1. Storage and image-only decoding

An encoded matrix is converted to an 8-bit image through the storage rule $\mathcal{Q}_m$. GASF and GADF use the known range $[-1, 1]$, MTF uses $[0, 1]$, and XIRP uses $[1, 2]$ on the diagonal and $[-\log 2, \log 2]$ off it. For signed RP, the per-image affine range is not supplied to the decoder. Instead, the quantised diagonal identifies the zero level; subtracting its median recovers a signed matrix up to an unknown positive factor. The subsequent geometric steps and final range normalisation are invariant to this factor. All choices needed to decode the *normalised* trajectory are therefore made from the stored image and globally fixed method settings, not from the original window.

4.2. Geometric reconstruction of signed RP

The proposed decoder recovers state geometry from distances, finds the orientation compatible with consecutive overlapping states, and finally uses directional signs to resolve the remaining reflection. For a length-T signal, the procedure is:

1. *Distance recovery.* Re-centre the stored pixel matrix at its diagonal zero level and symmetrise the absolute values:
$$R^{\mathrm{obs}} = \mathrm{PNG} - \mathrm{median}_i\, \mathrm{PNG}_{ii}, \qquad D_{ij} = \frac{|R^{\mathrm{obs}}_{ij}| + |R^{\mathrm{obs}}_{ji}|}{2}.$$
2. *State geometry.* Apply classical multidimensional scaling [15]. With $J = I - \mathbf{1}\mathbf{1}^\top/(T-1)$, form $B = -\frac{1}{2} J D^{\circ 2} J$. Its two leading eigenpairs give $Y = V_2 \Lambda_2^{1/2}$, a state configuration determined up to an orthogonal transformation.

3. *Overlap-compatible orientation.* Adjacent states must share one sample: for oriented rows (a_i, b_i), ideally $b_i = a_{i+1}$. Parameterise the unknown orthogonal transform by $Q = \mathrm{rot}(\theta)\,\mathrm{diag}(1, \eta)$, $\eta \in \{-1, +1\}$. For each η, the residual has the form $b_i - a_{i+1} = A_i \cos\theta + B_i \sin\theta$, with

$$A_i = \eta Y_{i2} - Y_{i+1,1}, \qquad B_i = -\eta Y_{i1} - Y_{i+1,2}.$$

   After removing the mean residual, the least-squares problem reduces to $\min_{\|u\|=1} u^\top H_\eta u$, where $u = (\cos\theta, \sin\theta)$ and $H_\eta = \tilde{C}^\top \tilde{C}$, $\tilde{C} = [\tilde{A}\ \tilde{B}]$. The smallest-eigenvalue eigenvector of each 2×2 matrix, taken with both signs, gives four candidates.

4. *Directional disambiguation.* Re-encode each candidate using the signed RP and select the one whose unit-norm matrix is closest to the unit-norm observed matrix in Frobenius norm. This comparison uses the image itself, never the original signal.

5. *Sample assembly.* For the selected candidate, let $\delta^\star = -\,\mathrm{mean}_i(b_i - a_{i+1})$ and correct $a_i \mapsto a_i - \delta^\star/2$, $b_i \mapsto b_i + \delta^\star/2$. Set $\hat{x}_1 = a_1$, $\hat{x}_i = (b_{i-1} + a_i)/2$ for $2 \le i \le T-1$, and $\hat{x}_T = b_{T-1}$. Finally map its range to the method's normalised domain.

The orientation problem is linear in T, while a dense eigendecomposition dominates the implementation cost, $O(T^3)$.

4.3. RP baselines and controlled comparisons

Geometric decoder without sign.. To isolate the information contributed by directional labels, we also apply the geometric procedure to unsigned distances. We retain the MDS, overlap-fitting and sample-assembly steps, but cannot

perform the signed-image comparison. On non-degenerate exact inputs, both orientations have the same distance image, so an image-only decoder cannot determine which one produced the original.

Hirata et al.. As a binary-RP reference, we implement the graph-distance and MDS construction of Hirata et al. [12, 23]. Thresholding gives a recurrence graph with edges $B_{ij} = 1$. For neighbour sets $G_i = \{k : B_{ik} = 1\}$, an edge receives weight $1 - |G_i \cap G_j|/|G_i \cup G_j|$. All-pairs shortest paths provide estimated distances, followed by MDS. Converting the resulting coordinates into a length-T series is our post-processing: retain the leading coordinate, append its mean, and rescale the range. We set the recurrence rate to 0.75 before benchmarking and report a sensitivity analysis over 0.02–0.75 (Section 6.2). When starting from a signed RP, the same unsigned graph determines the geometry; its directional signs are used only in an additional orientation-selection step. Thus this adaptation should not be confused with Hirata's original output procedure.

Sipers et al.. For continuous unsigned RP, we implement the spectral method of Sipers et al. [14]. It exploits the factorisation of two-dimensional Fourier coefficients of the squared distance matrix and reconstructs the signal up to a sign. The published construction assumes periodicity, so we provide it with a *circular* RP that includes the wraparound state (x_T, x_1). Image-derived frequency components below 10^{-4} of the strongest estimated power are discarded; the shared threshold was selected in a pilot and its effect is reported in Supplementary Section S5. Signed RP reconstruction adds our directional orientation-selection step. Because circularisation changes the encoded matrix, decoder comparisons are also performed on the *same circular*

image (Supplementary Table S5), rather than attributing an encoding change to Sipers' method.

4.4. Restricted-range GADF decoder

GADF provides angular differences but no absolute angle. Our procedure reconstructs a reference angle from the image and then recovers the remaining angles from relative differences. The method assumes the fixed normalisation $[-\alpha, \alpha]$ with $\alpha = 0.7 < 1/\sqrt{2}$.

1. *Image-derived boundary anchor.* Compute row means

$$\mu_i = \frac{1}{T} \sum_j G^{\mathrm{D}}_{ij} = \rho \sin(\phi_i - \psi),$$

   where $\rho \geq 0$ and ψ lies within the angular interval spanned by the samples. That interval is narrower than $\pi/2$, so the row means are monotone in ϕ_i. Thus $e = \arg\max_i |\mu_i|$ is an endpoint sample. Fix $\phi_e = \arccos(-\alpha)$ when $\mu_e > 0$ and $\phi_e = \arccos \alpha$ otherwise.

2. *Relative-angle recovery.* For pivot row p, recover every angle by

$$\hat{\phi}^{(p)}_j = \phi_e - \arcsin G^{\mathrm{D}}_{ep} - \arcsin G^{\mathrm{D}}_{pj}.$$

3. *Quantisation mitigation.* Average the angle estimates obtained from the five rows with the smallest $|\mu_p|$, then return $\hat{\hat{x}}_j = \cos\Big(\frac{1}{5} \sum_p \hat{\phi}^{(p)}_j\Big)$.

Under the normalisation assumptions the decoder is exact before storage and has $O(T^2)$ complexity. Pivot averaging is a numerical choice to mitigate pixel perturbations, not a condition for identifiability.

4.5. GASF, MTF and XIRP reference decoders

GASF.. We apply the established diagonal inverse on the $[0, 1]$ domain [1]. Its error after 8-bit storage can therefore be distinguished from the exact-input inverse.

MTF.. Since the image does not identify within-state amplitudes, our decoder constructs a representative sequence rather than an exact inverse. The T image rows are grouped into $Q = 4$ clusters; we estimate state-to-state transitions from the corresponding image blocks and order the clusters using the Fiedler vector of a symmetrised transition-graph Laplacian [16]. State k is assigned the bin midpoint $(k - \frac{1}{2})/Q$. Neither absolute state order nor within-state amplitudes are guaranteed by the image, so the resulting error combines irreversible encoding loss with approximation choices of the decoder.

XIRP.. We use the inversion-by-mean procedure of Hellermann and Lessmann [4]. Since the original normalised values appear on the matrix diagonal, this method serves as an invertible reference whose remaining error after quantisation can be measured directly.

5. Experimental protocol

5.1. Datasets and windows

Table 2 lists the seven public datasets: human activity (WISDM [17], PAMAP2 [25]), environmental audio (ESC-50 [26]), electrocardiography (PTB-XL [27]), geomagnetism (Swarm [28]), seismology (STEAD [29]) and surface electromyography (EMG [30]). Every channel is encoded, stored and decoded as its own single-channel image with its own calibration. Recordings are cut

Table 2: Datasets. Windows of 128 samples available after segmentation; the group is the unit that never appears on both sides of a split.

Dataset	f_s (Hz)	Channels	Task	Windows	Group
WISDM	20	3	5 activities	8 172	subject (51)
PAMAP2	100	9	12 activities	15 085	subject (9)
ESC-50	8 000	1	50 sound classes	624 000	source recording
PTB-XL	100	12	5 classes, multilabel	21 000	patient
Swarm MAG	1	3	none	62 757	–
STEAD	100	3	earthquake vs noise	92 000	event / station
EMG gestures	100	8	6 gestures	880	subject (36)

into non-overlapping windows of $T \in \{64, 128, 256\}$ samples that never cross a label change or a time gap. Blocks with fewer than 200 windows (EMG at $T = 256$) are not reported.

5.2. Fidelity, robustness and cost

Fidelity uses 500 windows per dataset and length drawn with a fixed seed, and 3 000 for the single-channel ESC-50, so that every block holds between about 1 500 and 6 000 channel series. Writing $r = \max_i x_i - \min_i x_i$, it is measured per channel series by

$$\mathrm{NMAE} = \frac{1}{rT} \sum_i |x_i - \hat{x}_i|,$$

the Pearson correlation, the L_1 distance between unit-mass Welch spectra [18] and the relative energy error; constant series are excluded. Each decoder is run on the floating-point matrix and on the 8-bit image. Robustness adds zero-mean Gaussian noise of $\sigma \in \{2, 4, 8, 16\}$ intensity levels to every pixel of the stored image, which is then clipped and rounded, with two replicates per

level on 200 windows per dataset and length. Cost is the single-thread time to encode, store and decode and the peak traced memory, for T from 64 to 1 024. The sampling-rate study reads 3 000 PTB-XL records at 500 Hz and resamples them with an anti-aliased polyphase filter to 250, 100 and 50 Hz, with windows of 128 samples and, as a control, of a fixed duration of 1.28 s.

5.3. Classification utility

Classifiers are trained on original windows and evaluated on the original and on every reconstruction of the same held-out windows. Up to 6 000 windows per dataset and length are split into three grouped folds that keep every subject, patient, event, station or source recording on one side, stratified by class for single-label data; per-channel standardisation uses training statistics only, and reconstructions are produced independently per window, so no information crosses the split. KNN, an RBF SVM, a random forest, an LSTM, a GRU and a one-dimensional CNN are used with standard settings (Supplementary Section S1). Accuracy, or macro-F1 for the multilabel PTB-XL, is averaged over the three folds. Swarm has no labels. ESC-50 windows of 128 samples span 16 ms of sound, which leaves every classifier near chance, so audio is evaluated on the official ESC-10 subset at 2, 4 and 8 kHz with windows of 512 and 1 024 samples, silent windows removed and accuracy scored per clip.

5.4. Statistical protocol

All comparisons are paired over the same series; differences between decoders are summarised by medians of per-series differences, averaged across datasets with equal weight. A reconstruction is declared *equivalent* to the

original for a classifier when the 95% bootstrap interval of the accuracy difference, resampling whole groups [19, 20], lies within ± 0.01, which corresponds to two one-sided tests at the 2.5% level [22, 21]; a non-significant difference is not taken as evidence of equivalence. Equivalence is assessed on single-label data; for PTB-XL only the drop in macro-F1 is reported.

6. Results

6.1. Fidelity and storage (RQ1)

Table 3 gives the 8-bit error of every combination at $T = 128$; the other lengths behave alike (Supplementary Tables S1–S2). Over the 20 reported dataset–length blocks, 61 079 channel series were reconstructed by every combination without decoder failure.

On the floating-point matrix, the geometric decoder on the signed RP, GASF, GADF and XIRP recover the series to machine precision: the median NMAE per block never exceeds 1.4e-13 for the signed RP and 5.6e-17, 8.4e-17 and 2.8e-16 for GASF, GADF and XIRP. Over every series of every block, the largest error is 1.6e-11 for the signed RP and 8.0e-13, 3.6e-16 and 6.3e-16 for GASF, GADF and XIRP. For these four, the 8-bit error is therefore storage error. It stays between 0.0009 and 0.0022 for the signed RP on six datasets and rises to 0.012–0.017 on Swarm; GASF and GADF are of the same order, and XIRP is about an order of magnitude lower. Sipers' reconstruction on the circular signed RP comes close, between 0.0028 and 0.0155. Without a node threshold it is exact in floating point but less accurate after storage; with the threshold used here its floating-point error reaches a median of 1.7e-03 per block (Supplementary Section S5).

Table 3: Mean 8-bit NMAE of every encoding–decoder combination at $T = 128$ (500 windows per dataset, 3 000 for ESC-50; every channel). Uns./Sig.: unsigned and signed RP.

	Hirata		Sipers		Geometric (ours)					
Dataset	Uns.	Sig.	Uns.	Sig.	Uns.	Sig.	GASF	GADF	MTF	XIRP
WISDM	0.2411	0.1044	0.1954	0.0043	0.1946	0.0010	0.0015	0.0013	0.2838	0.0002
PAMAP2	0.2345	0.1059	0.1884	0.0050	0.1899	0.0015	0.0016	0.0013	0.2771	0.0002
ESC-50	0.2184	0.0840	0.1916	0.0059	0.1922	0.0017	0.0016	0.0011	0.2740	0.0002
PTB-XL	0.3313	0.1982	0.2680	0.0055	0.2874	0.0016	0.0029	0.0023	0.3404	0.0002
Swarm	0.2778	0.0283	0.2837	0.0045	0.2547	0.0124	0.0026	0.0015	0.2979	0.0002
STEAD	0.2044	0.0868	0.1802	0.0051	0.1771	0.0013	0.0015	0.0011	0.2668	0.0002
EMG	0.1979	0.1205	0.1669	0.0039	0.1680	0.0009	0.0013	0.0022	0.2834	0.0006

The remaining combinations reach an NMAE between about 0.03 and 0.34, for two different reasons (Figure 3; spectral and correlational metrics in Supplementary Table S7). On the unsigned RP, the geometric and Sipers' decoders keep the spectrum as well as on the signed RP, because a reflection leaves the spectrum unchanged; its pointwise error and its correlation of about zero come from returning the reflection in about half of the series. Hirata's reconstruction and the MTF representative also distort the spectrum and the energy: they lose the waveform, not only its sign.

6.2. Recurrence plots: encoding versus decoder (RQ2)

Table 3 separates five effects. The spectrum is unchanged by a reflection of the series, so it shows what each decoder keeps of the waveform independently of its orientation. *Sign.* On the unsigned RP, the geometric decoder keeps the spectrum as well as on the signed RP (PSD-shape distance 0.0007–0.0151, against 0.0034–0.0187), while its correlation with the original stays within ± 0.05: it recovers the shape and loses only the orientation, which



Figure 3: The first channel of the windows of Figure 2, decoded from its 8-bit image. Left: the geometric decoder on the signed and on the unsigned RP. Right: Hirata's reconstruction on the unsigned RP and MTF. GASF, GADF and XIRP overlap the original at this scale and are omitted. On Swarm, whose states are nearly collinear, the signed RP deviates visibly at the start of the window.

the sign restores from the image alone (Figure 3). *Thresholding.* Hirata's reconstruction from the binarised RP distorts the spectrum itself (0.06–0.61): once the image is binary, the waveform is lost, not only its sign. *Continuous distances.* On the same signed RP, the geometric decoder, which uses the distances that thresholding discards, is one to two orders of magnitude more accurate than Hirata's on six datasets; on Swarm the gap narrows to a factor of about two. *Decoder on continuous distances.* Sipers' reconstruction also uses the continuous distances; with the sign it is a median of 0.0015 per series above the geometric decoder, which on the same circular image is 1.8 to 4.6 times more accurate on every dataset (Supplementary Table S5). It is ahead in the benchmark only on Swarm, and the gain comes from the encoding: adding the wrap state raises the median σ_2/σ_1 from 0.0003 to 0.15 and lowers the geometric decoder's error from 0.0122 to 0.0025, below Sipers' 0.0046 on the same image, while on other datasets it changes by at most 0.0002. *Threshold choice.* These comparisons do not depend on the recurrence rate fixed for Hirata: choosing the best rate per dataset with knowledge of the result lowers its error by at most 0.025 on the unsigned and 0.010 on the signed RP, leaving it between 0.02 and 0.31 (Supplementary Table S3).

6.3. Signal characteristics and stability (RQ3)

Storage error is expected to be amplified when the states are nearly collinear, because classical MDS then recovers a two-dimensional subspace separated by a small eigengap [24], and the data agree. None of the 54 040 series with $\sigma_2/\sigma_1 \geq 0.1$ exceeds an 8-bit NMAE of 0.01, whereas 85% of the series with $\sigma_2/\sigma_1 < 0.01$ do. The ratio is an empirical indicator, not a complete explanation: it separates the regimes at the two ends but does not

determine which series in between fail. Nor does the share of sign-informative pairs: slow series are informative in nearly every pair, yet their states are the most collinear. Swarm, sampled at 1 Hz, is the limiting case and the only dataset where the signed RP is not in the low-error group.

The sampling rate changes how far apart consecutive samples are, and with it the conditioning (Figure 4b–c). On PTB-XL with windows of 128 samples, from 50 to 500 Hz the median σ_2/σ_1 falls from 0.61 to 0.08, the share of series below 0.1 grows from 0% to 67%, and the 8-bit NMAE of the signed RP rises from 0.0009 to 0.0076, while GASF and GADF stay between 0.0018 and 0.0032 and XIRP is unaffected. A fixed number of samples spans a shorter time at a higher rate, but the trend persists at a fixed duration of 1.28 s, from 0.0012 to 0.0079 (Supplementary Table S4). Sipers' reconstruction rises with the rate as well (Figure 4b), so the effect belongs to the RP, not to one decoder. The resampled and the distributed 100 Hz files give the same error (0.0015 and 0.0015). Resampling also changes the spectral content of each window, and the study covers one dataset, so the trend is attributable to the rate as realised by anti-aliased resampling of PTB-XL, not to the rate in isolation.

6.4. Classification utility (RQ4)

Six classifiers, five labelled datasets and three lengths give 84 combinations on 72 917 windows (Table 4). On the original windows, accuracy ranges from 0.17 to 0.81 and PTB-XL macro-F1 from 0.14 to 0.61; the lowest values (KNN on EMG) are near chance, where drops are less informative. The four combinations that are exact in floating point never lose more than 0.007 and are equivalent within ± 0.01 in nearly every single-label combination; where

Table 4: Drop from the original accuracy (macro-F1 on PTB-XL) for classifiers trained on original windows and tested on reconstructions. $|\Delta| \leq 0.01$: share of combinations whose observed drop is within 0.01. Equivalent: 95% group-bootstrap interval within ± 0.01, over the single-label combinations.

	Five datasets, $T \in \{64, 128, 256\}$				ESC-10, per clip	
Decoder	Mean drop	Max drop	$\lvert\Delta\rvert \leq 0.01$	Equivalent	Mean drop	Equivalent
Hirata / uns.	0.120	0.376	19%	11/66	0.071	0/36
Hirata / sig.	0.080	0.283	26%	12/66	0.060	0/36
Sipers / uns.	0.098	0.371	19%	13/66	0.021	0/36
Sipers / sig.	0.000	0.008	100%	59/66	0.001	28/36
Ours / uns.	0.099	0.359	21%	13/66	0.024	0/36
Ours / sig.	0.000	0.007	100%	63/66	0.001	33/36
GASF	-0.000	0.005	100%	63/66	0.001	34/36
GADF	-0.000	0.002	100%	63/66	0.000	32/36
MTF	0.099	0.400	19%	7/66	0.106	0/36
XIRP	0.000	0.002	100%	65/66	-0.000	36/36

equivalence is not established, the observed difference is still below 0.01 and the interval is too wide rather than the drop large. Sipers' reconstruction with the sign joins this group, with a largest drop of 0.008, but is equivalent in fewer combinations (59 of 66, against 63 for the geometric decoder), consistent with its larger 8-bit error. Without the sign, Sipers' reconstruction, Hirata's on either encoding, the geometric decoder on the unsigned RP and MTF lose between 0.08 and 0.12 on average and up to 0.40: whichever decoder is used, the unsigned RP loses the orientation and with it the accuracy. The same separation holds per clip on ESC-10 (18 000 windows, 36 combinations), where the exactly decodable combinations lose at most 0.017.

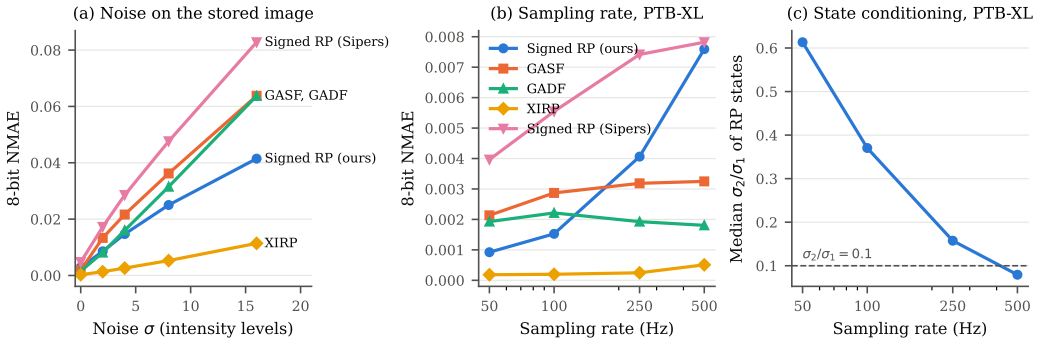


Figure 4: (a) Mean 8-bit NMAE against Gaussian noise added to the stored image ($T = 128$, mean over the seven datasets, two replicates per level). (b) 8-bit NMAE and (c) median conditioning σ_2/σ_1 of the RP states against the sampling rate (PTB-XL, $T = 128$, anti-aliased resampling from 500 Hz). Hirata's reconstruction, the geometric and Sipers' decoders on the unsigned RP and MTF stay between 0.08 and 0.34 throughout and are omitted.

6.5. Robustness (RQ3) and cost (RQ4)

With noise of $\sigma = 16$ intensity levels on the stored image, the mean NMAE at $T = 128$ rises to 0.011 for XIRP, 0.041 for the signed RP, and 0.064 and 0.064 for GASF and GADF, and 0.083 for Sipers' reconstruction on the signed RP (Figure 4a). The geometric decoder on the signed RP degrades less than both angular fields on 6 of the 7 datasets, with the three level on Swarm, and less than Sipers' on 7 of the 7. The lossy combinations move by at most 0.020, because their error is already dominated by what is discarded before storage.

Hirata's reconstruction has the steepest cost, with an exponent of about 2.6 due to all-pairs shortest paths (Supplementary Table S6). The geometric decoder on the signed RP is 2.9 times faster at $T = 256$ and 5.2 times faster at $T = 1024$, and costs about as much as Sipers' reconstruction at $T = 256$ and half as much at $T = 1024$. GASF, GADF, XIRP and MTF are one to two

orders of magnitude cheaper. Memory grows quadratically for all of them.

7. Discussion

What each representation retains. Three groups emerge, consistent across domains, lengths and classifiers. The signed RP with the geometric decoder, GASF, GADF on its restricted range and XIRP are exact in floating point under the stated conditions, so their error after storage is storage error, too small to affect classifiers trained on original signals. The unsigned RP determines the shape of the series but not its sign. The binary RP and MTF are many-to-one, so no decoder can be exact for every series, and the evaluated decoders lose most of the waveform; better approximations, for instance with learned priors, are not ruled out, but they cannot be read back with a guarantee of matching the encoded series.

What the sign and the decoder add. RP reconstruction needs both. The sign fixes the orientation that the unsigned RP leaves open, and one sign-informative pair suffices (Section 3). The continuous distances that thresholding discards account for the advantage over Hirata's reconstruction, and among decoders that use them the geometric one is more accurate and more robust than Sipers' on the same image, without assuming a periodic signal or choosing a frequency threshold.

When recovery becomes unstable. The signed RP is the only exactly decodable representation whose stability depends strongly on the signal: slowly varying windows, from Swarm at 1 Hz or from an ECG at 500 Hz, have nearly collinear states that amplify storage error, for Sipers' decoder as for ours.

Using the benchmark. The comparison is not a ranking, since a model may need a particular image. With recurrence plots, the signed RP and the geometric decoder give a route back to the series, with sufficient conditions for exactness, a measurable indicator of stability and less sensitivity to pixel noise than the angular fields on most datasets. GADF can be decoded if its angular range is restricted at encoding time; MTF and the binary RP give only approximations. Where the image is free to choose and the signal is sampled far above its bandwidth, the angular fields and XIRP are cheaper and independent of its conditioning. XIRP is an explicitly invertible reference: the redundancy of its column proposals plausibly explains its lower error, which says nothing about whether its images are better inputs for classification or generation.

Limitations. Images are deterministic outputs of known encoders; pixel noise measures local stability only. Physical units need the per-channel calibration, which a generative setting would have to supply; only 8-bit storage and independent channels are studied. PTB-XL and ESC-50 are split by patient and recording, not by official folds. Hirata's reconstruction is our implementation, with a threshold the method does not prescribe, and Sipers' runs on the circular RP it requires, with a node threshold chosen in a pilot. The sampling-rate study covers one dataset, and σ_2/σ_1 is an empirical indicator, not an error bound.

8. Conclusions

We measured, under one framework, how much of a time series survives its image for the most widely used image representations and XIRP, on seven

datasets from six domains. Four are exactly decodable in floating point under stated conditions and, after 8-bit storage, keep what classifiers trained on original signals need: the signed recurrence plot with a geometric decoder, GASF on $[0, 1]$, GADF on a restricted angular range, and XIRP. The unsigned recurrence plot is recovered only up to sign; the binary recurrence plot and the Markov transition field, which are many-to-one, lose most of the waveform with the decoders evaluated. For recurrence plots, exact recovery needs the sign and a decoder that uses the continuous distances; stability after storage depends on how collinear the states are, and hence on the sampling rate. Images produced by generative models are the next step.

CRediT authorship contribution statement

Adriano García-Giralda Milena: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Francisco García Moreno:** Conceptualization, Methodology, Supervision, Writing – review & editing. **José Antonio Jiménez Madrid:** Methodology, Formal analysis, Writing – review & editing. **María Bermúdez Edo:** Supervision, Writing – review & editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

All datasets are public and cited. Code, manifests, configurations and derived results: `https://github.com/adrianoggm/TFG-Tratamiento-de-Series-Temporales` [pending: DOI, #137].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[pending: tools and uses, #138.] All AI-assisted content was critically reviewed, verified and revised by the authors, who take full responsibility for the accuracy, originality and integrity of the manuscript.

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